



PROJECT AGRIVISION

Machine Learning for Smallholder Agribusiness

An applied ML internship: early crop-disease detection, from strategic planning through data engineering, model development, and field deployment.

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What this presentation covers



01

The Problem

Why crop disease diagnosis is slow, and what it costs smallholder farmers



02

The Solution

AgriVision — a mobile-first ML diagnostic tool and its architecture



03

Four-Week Journey

Strategy → Data → Model → Deployment, week by week



04

Outcomes & Next Steps

Success metrics, risks, and the path to scale

CONTEXT

About the internship



Organization

YuvaIntern, in association with Henry Harvin Education (NSDC-aligned)



Role

Junior Machine Learning Engineer – Agribusiness



Duration

September 10 – October 08, 2026 (4 weeks, remote)



Initiative

Internally codenamed “Project AgriVision”



Why it matters

Smallholder farmers grow most of the food in many developing regions, yet often wait a week or more for an expert to confirm a crop disease diagnosis. By the time treatment arrives, low-cost intervention is no longer effective.

This internship worked through a realistic, deployable answer to that gap — an ML tool a farmer can reach directly from a phone.

THE PROBLEM

Diagnostic delay costs the harvest

Smallholder farmers detect disease through visual inspection and word-of-mouth, then travel to or wait for an extension officer to confirm it. The process is slow, inconsistent across regions, and doesn't scale during an outbreak.

5–7 days

Typical wait for a confirmed diagnosis today

1/3–1/2

Of potential harvest lost to pests/disease in affected sub-Saharan African crops

\$100s of billions

Estimated annual global economic cost of crop pathogens

Why machine learning fits

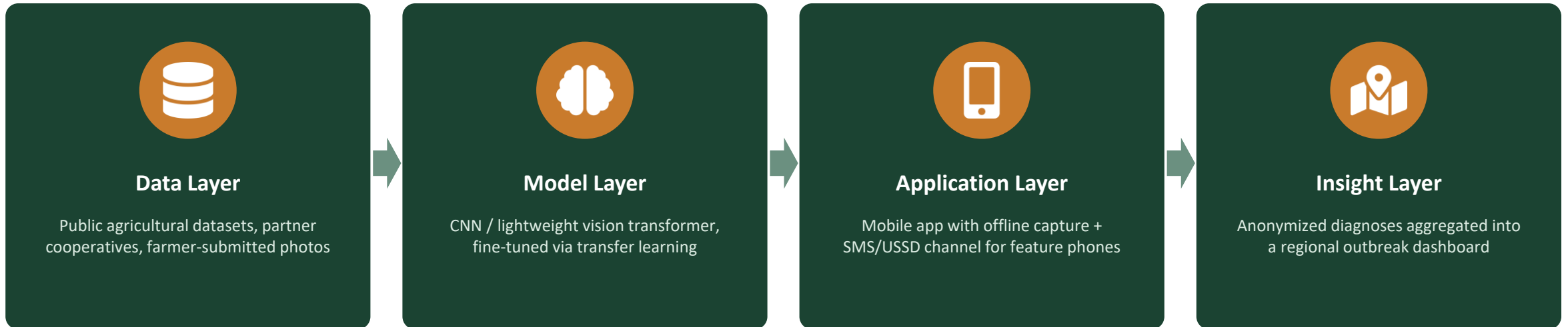
- Visual symptoms (leaf spots, wilting) suit image classification, which has matched expert-level accuracy in published studies
- A trained model diagnoses in seconds, on-device or via low-bandwidth server call
- Marginal cost per diagnosis is near zero once trained — scales to thousands of farmers
- Aggregated, anonymized results can power a live regional disease-outbreak map

GOALS

Project objectives (SMART-scoped)

01	Model accuracy	Image classifier for the top 5 pilot-crop diseases	≥ 90% top-1 validation accuracy
02	Response speed	Mobile / USSD photo-based diagnosis, usable offline	< 15 seconds on a mid-range phone
03	Turnaround time	Cut the symptom-to-diagnosis gap for pilot farmers	~5–7 days → under 24 hours
04	Real-world impact	Validate yield & pesticide-use impact vs. control group	Measurable reduction in yield loss
05	Continuous learning	Data pipeline & feedback loop for retraining	Automated pipeline live before pilot end

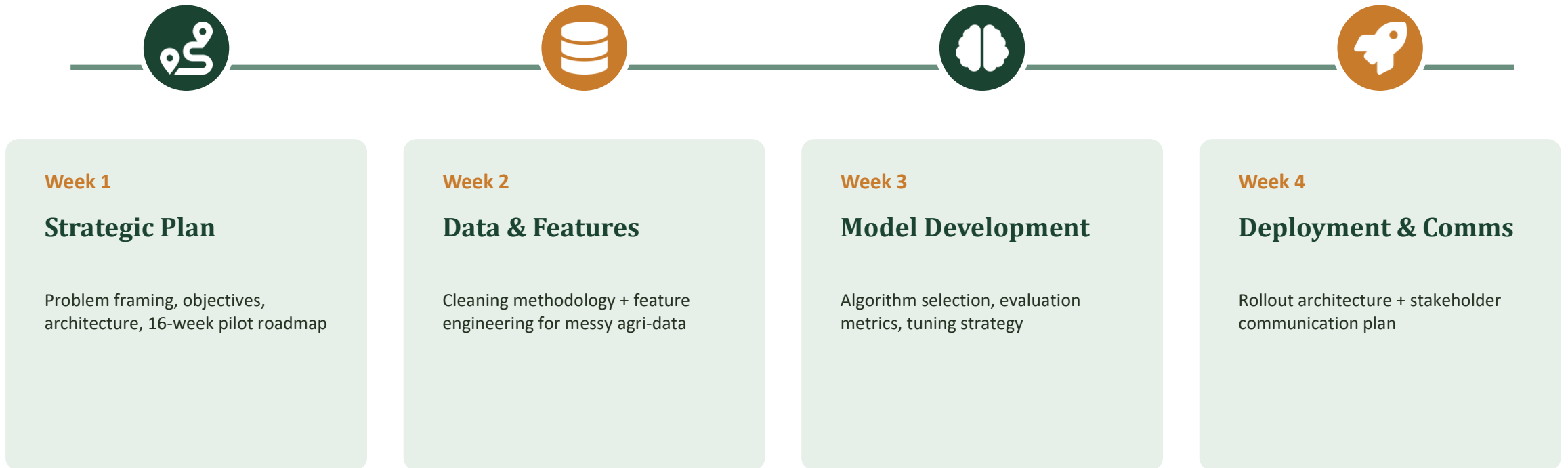
Four-layer solution architecture



Deployment model: hybrid cloud-edge

- Containerized model on a managed cloud endpoint, autoscaled for seasonal demand
- Lightweight mobile / progressive web app with offline caching of the last recommendation
- SMS / USSD / WhatsApp fallback for smallholder farmers without smartphones or data plans

A four-week, deliverable-driven journey



Agricultural data is messy — on purpose-aware terms



Guiding principle: data-quality issues follow agronomic logic (sensors fail in storms, farmers skip readings at harvest) — fixes should follow domain knowledge, not generic statistics.



Missing Data

Sensor downtime, unreported harvests

Time-aware interpolation, KNN spatial imputation, multiple imputation with flags



Outliers

Entry errors vs. real droughts/floods

Region & crop-specific thresholds; cap true errors, flag & keep real extremes



Scale Mismatch

mm rainfall, tons yield, pH, ppm nutrients

Min-max, z-score, and log transforms — scaled per crop / agro-ecological zone



Engineered Features

Raw variables under-inform on their own

Growing Degree Days, N:P:K ratios, water balance index, NDVI trajectory

Matching algorithms to agribusiness constraints

Chosen against three criteria: interpretability (can an agronomist trust it), speed, and scalability.

Algorithm	Best Use	Interpretability	Speed
Linear / Logistic Regression	Yield trend estimation, simple risk scoring	High	Very fast
Random Forest	Yield prediction, disease-risk classification	Moderate	Moderate
Gradient Boosting (XGBoost/LightGBM)	High-accuracy yield & price forecasting	Moderate (+SHAP)	Fast
CNN (Transfer Learning)	Image-based disease/pest detection	Low (+Grad-CAM)	Fast to infer
Time-Series (ARIMA/Prophet/LSTM)	Seasonal yield & price forecasting	Moderate	Moderate

Recommended approach: Gradient Boosting for tabular yield/price problems (paired with SHAP for explainability), Random Forest as a robust fallback, and a transfer-learning CNN for image-based disease detection.

Metrics that reflect real agricultural stakes

Standard ML metrics alone hide the cost of being wrong: a missed outbreak (recall failure) can cost a season's harvest; a false alarm (precision failure) wastes scarce inputs and erodes trust.

Recall (Sensitivity)

Missed disease/pest cases

Critical — a miss can devastate a harvest

Precision

False positive treatments

High precision avoids wasted chemicals & cost

MAE / RMSE

Yield prediction error (tons/ha)

Reported in farmer-meaningful units

Actionability Lead Time

Days of advance warning

Ties model output to a farmer's ability to act

The model only helps if people trust it



Smallholder Farmers

“Will this help me grow more, and is it easy to use?”

SMS, voice/USSD, local-language demos



Agronomists

“Can I trust this and explain it to farmers?”

Field workshops, one-page reference guides



Cooperative Managers

“What does this cost, and how do I roll it out?”

ROI one-pager, dashboard demo



Business Sponsors

“What is the return on investment?”

Executive summary, quarterly KPI briefing

Key risks and how they're mitigated

	Low connectivity blocks access	High	Offline caching + SMS/USSD fallback channel
	Low farmer trust / adoption	High	Local champions, co-design pilot, transparent overrides
	Model drift across seasons	Medium	Scheduled drift monitoring & retraining cycle
	Data privacy concerns	Medium	Anonymized aggregation, informed consent, clear data policy

What success looks like at pilot's end



Model Quality

Validation accuracy,
precision/recall per disease
class



System Performance

On-device inference speed,
stability under low connectivity



Operational Impact

Reduced diagnosis turnaround,
lower disease-related yield loss



Adoption

Active farmers, repeat usage,
onboarding feedback



Business Impact

Reduced input waste, scalable
cost per new region

Primary pilot KPI: reduced diagnosis turnaround time and a measurable decrease in disease-driven yield loss, versus control plots, across a 16-week pilot.

KEY TAKEAWAYS

From strategy to a deployable playbook

- A technically strong model fails if it ignores field connectivity, hardware, and trust — deployment planning matters as much as model accuracy.
- Agricultural data quality issues follow agronomic logic, not random noise — domain knowledge should drive every cleaning and feature-engineering decision.
- Recall and actionability lead-time matter more than raw accuracy when a missed outbreak can cost an entire harvest.
- The four-week arc — strategy, data, model, deployment — produced a working playbook a real team could execute against.



Thank You

Questions and discussion welcome.

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